

1. A 25-year-old man is admitted to a medical ward with symptoms of irritability and lethargy. The physician working in the ward did a lumbar puncture, and found CSF glucose 20 mg/dL, Protein 200 mg/dL, Opening pressure 250 mm CSF, White blood cell count 1600 cells/ μ L with 92% Polymorphonuclear Leukocytes and the CSF red blood cell (RBC) count 50/ μ L. Lastly, the clinical team set the diagnosis to be Meningitis.

What is the most likely cause of the meningitis in this patient?

- (A) Viral
- (B) Fungal
- (C) Bacterial
- (D) Tuberculosis

Answer: C

2. A 25-year-old woman with no previous history of UTI, complains of burning on urination, frequent but small amount of urine, and bladder pain. She has no fever or costo-vertebral angle tenderness. A clean-catch midstream urine sample shows gram-negative rods on Gram stain. Based on these findings, she is presumed to have a lower UTI.

What is the most likely bacterial etiology in this patient?

- (A) *Proteus mirabilis*
- (B) *Pseudomonas aeruginosa*
- (C) *Escherichia coli*
- (D) *Klebsiella pneumonia*

Answer: C

3. A 32-year-old man with HIV infection comes to an ART clinic. The physician in charge was considering to prescribe Tenofovir based fixed drug combination but doubted whether prior investigations need to be done ahead or not and consulted the pharmacist on duty.

What laboratory test does the pharmacist recommend in this case?

- (A) Electrocardiographic test
- (B) Complete blood count
- (C) Liver function test
- (D) Renal function test

Answer: D

4. A 27-year-old man was brought to an emergency department for complaints of nuchal rigidity, fever and altered mental status of half a day duration. The physician in charge well investigated and set his diagnosis as acute bacterial meningitis and planned to initiate adjunctive dexamethasone therapy with antibiotics. However, the physician hesitated with the appropriate timing of Dexamethasone therapy; hence, requested the clinical pharmacist on duty to support with this regard?

What is the most recommended Dexamethasone initiation time for this patient?

- (A) Initiate 20 minutes after the first dose of antimicrobials for 2 days
- (B) Initiate 20 minutes after the first dose of antimicrobials for 2 weeks
- (C) Initiate concomitant with the first dose of antimicrobials for 2 weeks
- (D) Initiate 20 minutes prior to the first dose of antimicrobials for 2 days

Answer: D

5. A 22-year-old known type II diabetic patient came to an OPD with complaints of productive cough, night sweats and weight loss of two weeks duration. Upon investigation, it was evidenced that the sputum was positive for AFB and the clinical team concluded the diagnosis as pulmonary tuberculosis; hence, put the patient on (Isoniazid 300mg daily, Rifampicin 600mg daily, Pyrazinamide 1600mg daily and Ethambutol 1100mg daily) for 2 months. Otherwise, the plasma glucose was well controlled with minimal complaints against the respective oral hypoglycemic agents he was on.

What is the most likely treatment problem occurred in this patient?

- (A) Dosage too low
- (B) Dosage too high
- (C) Drug non-adherence
- (D) Needs additional drug therapy

Answer: D

6. A 40-year-old man wants to go to a malaria endemic area for two weeks. The most common plasmodium species in the area is *P. falciparum*. He visited a community pharmacy, and requested the pharmacist for anti-malarial prophylaxis before his travel.

What is the most appropriate drug for this patient?

- (A) Doxycycline 100mg PO daily 2 days prior to travel, continued for 2 weeks after his return
- (B) Mefloquine 250 mg PO weekly 2 weeks prior to travel, continued for 1 week after his return
- (C) Primaquine 15mg PO daily 2 days prior to travel, continued for 2 weeks after his return
- (D) Mefloquine 250 mg PO weekly 2 weeks prior to travel, continued for 4 weeks after his return

Answer: D

7. A 34-year-old man, who has perforated peptic ulcer, comes with severe abdominal pain and nausea. His vital signs were: Temperature is 39.3°C and PR is 105 bpm. On investigation, he has a WBC count of $16.5 \times 10^3/\mu\text{L}$ and blood urea nitrogen of 34 mg/dL; and hence, diagnosed with generalized peritonitis secondary to gastric perforation. The intra-abdominal infection was managed by prompt drainage, fluid resuscitation and appropriate antimicrobial therapy.

What is the most effective monitoring parameter for this condition?

- (A) Urine output
- (B) Mental Status
- (C) Patient weight
- (D) Blood Pressure

Answer: A

8. A 30-year-old woman was diagnosed with mania and started a lithium therapy. One month later, the patient symptoms were improved but she started to experience some new symptoms. The physician suspects this could be due to drug side effect, so ordered laboratory tests.

What laboratory changes are expected in this patient?

- (A) Increased in thyroid hormones
- (B) Decreased in kidney function
- (C) Decreased in white blood cells
- (D) Increased in liver function tests

Answer: B

9. A 40-year-old man was prescribed with Haloperidol to treat schizophrenia. Later on, the physician changed the treatment to Risperidone since haloperidol therapy was unable to produce adequate clinical response. However, Risperidone trial also failed to produce an adequate clinical response.

What is the next most effective drug therapy for this patient?

- (A) Clozapine
- (B) Olanzapine
- (C) Thioridazine
- (D) Fluphenazine

Answer: A

10. A 22-year-old man on Fluoxetine 40mg tablets for the management of major depressive disorder for 8 weeks decided to discontinue medication. He reports that "I feel like I have my old life back". His energy has improved and he is doing well in his studies.

What is the next most appropriate step in the treatment of this patient?

- (A) Reduce Fluoxetine dose to 10mg
- (B) Continue Fluoxetine 40mg
- (C) Change to Sertraline 50mg
- (D) Add Imipramine 25mg

Answer: B

11. A 21-year-old female patient with chief complaints of depressed mood, loss of pleasure, and hypersomnia was admitted to inpatient psychiatry clinic. She also has history of a profound lethargy, hyperphagia, pathological sensitivity to interpersonal rejection for the last two

weeks. After full psychiatric assessment, the patient was diagnosed with atypical type of Major Depressive Disorder.

What is the most appropriate drug to be used in the management of this case?

- (A) Phenelzine
- (B) Paroxetine
- (C) Venlafaxine
- (D) Imipramine

Answer: A

12. A 25-year-old man came to a psychiatry clinic with complaints of poor concentration, recurrent thoughts of suicide, diminished interest, loss of energy, changes in sleep pattern and feelings of worthlessness of three weeks duration. The physician diagnosed the case as major depressive disorder but consulted the pharmacist in charge before initiating the drug therapy.

What is the most likely information the pharmacist should quest before his recommendation?

- (A) Depressed mood
- (B) Previous drug use
- (C) Change in appetite
- (D) Psychomotor agitation

Answer: B

13. A 50-year-old man is admitted to a hospital with an acute onset of severe mid-thoracic spinal pain. He is found to be anemic and investigations show that he has multiple myeloma with widespread bony lesions, including fresh spinal fractures.

What is the most appropriate pain management for this patient?

- (A) Ibuprofen
- (B) Tramadol
- (C) Acetaminophen
- (D) Morphine

Answer: D

14. A 38-year-old known epileptic woman is taking phenytoin 400 mg/day and Valproate 2,000 mg/day. She also uses high dose of oral contraceptives to prevent pregnancy. In addition, she is taking Methyldopa 250mg ²⁰ BID for her long standing hypertension. Today she came to the ambulatory clinic with complaints of painful lower limb swelling which later has been diagnosed as deep venous thrombosis.

What is the most likely medication responsible for her complication?

- (A) Phenytoin
- (B) Valproate
- (C) Methyldopa
- (D) Oral contraceptive

Answer: D

15. A 48-year-old patient with epilepsy came to neurology clinic for follow up. The patient seizure was well controlled for the last three years with phenytoin 100 mg BID. The neurologic exam and electroencephalogram (EEG) results are normal. The attending physician planned to stop the medication as the patient is seizure free for the last three years. He asked the clinical pharmacist working in the neurology clinic to suggest on the antiepileptic medicines discontinuation plan for this patient.

What could be the appropriate suggestion expected from the pharmacist in this case?

- (A) Do not stop the medication since it should be taken for life time.
- (B) Do not stop the medication since phenytoin has no adverse effects.
- (C) Stop the medication by tapering over a period of three months.
- (D) Stop the medication at once since the patient is currently seizure free.

Answer: C

16. A 59-year-old man was presented to a hospital with complaints of muscle weakness, fatigue and confusion. He has a two year history of hypertension treated with thiazide diuretic. He regularly drinks alcohol. His blood pressure is 107/70 mmHg. On investigation, laboratory results are Potassium: 3.2 mEq/L; RBC count: $4.2 \times 10^{12}/L$; Hgb: 13.6 g/dL; fasting Glucose, 112 mg/dL. (normal value: HGB=13.3–16.2 g/dL, Potassium=3.5–5.0 meq/L, RBC: $4.30-5.60 \times 10^{12}/L$, Fasting glucose: 75–100 mg/dL)

What is the most likely cause for the patient's condition?

- (A) Iron-deficiency Anemia
- (B) Postural hypotension
- (C) Alcohol intoxication
- (D) Adverse drug effect

Answer: D

17. A 48-year-old man with Stage C heart failure has been treated with Captopril 50mg PO TID, Bisoprolol 10mg PO daily, and Digoxin 0.25mg PO daily. Past history was remarkable for calcium Nephrolithiasis and currently treated with Hydrochlorothiazide 25mg PO daily. He visits the chronic care clinic with complaints of dizziness, light-headedness and blurred vision. Other findings are unremarkable.

What is the most likely drug therapy problem this patient faces?

- (A) Dosage too high
- (B) Dosage too low
- (C) Adverse drug reaction
- (D) Needs additional drug therapy

Answer: C

18. A 55-year-old woman comes to a physician with complaints of troubling headache and constipation. She was recently diagnosed with stable angina and being on treatment for two weeks.

What is the most likely drug that the woman was taking?

- (A) Nifedipine
- (B) Nitroglycerin
- (C) Verapamil
- (D) Propranolol

Answer: C

19. A 50-year-old man was recently diagnosed with hypertension. He has a two year history of chronic kidney disease. Appropriate drug therapy is instituted.

What is the high-priority blood pressure treatment goal in this patient?

- (A) <150/90 mm Hg
- (B) <140/90 mm Hg
- (C) <130/80 mm Hg
- (D) <120/80 mm Hg

Answer: B

20. A 54-year-old woman, who was recently diagnosed with Stage C heart failure and started low dose Aspirin, Captopril, Metoprolol and Digoxin; was found to have a third-degree atrioventricular block. Up on review of the past medical records, the ECG report dictates normal conduction. The drug therapy has adequately managed the patient's symptoms; however, the physician fears that the new condition was due to the medications being used.

What is the most likely intervention to be applied for this patient?

- (A) Discontinue Metoprolol
- (B) Discontinue Captopril
- (C) Discontinue Digoxin
- (D) Discontinue Aspirin

Answer: C

21. A 60-year-old woman has recently been diagnosed with Stage 1 hypertension and Stage B heart failure. The physician decided to initiate two drug therapies with the advisable life change modification.

What is the most appropriate drug therapy for this patient?

- (A) Hydrochlorothiazide and Nifedipine
- (B) Furosemide and Nifedipine
- (C) Captopril and Metoprolol
- (D) Furosemide and Captopril

Answer: C

22. A 54-year-old female has recently been diagnosed with chronic kidney disease has been diagnosed with hypertension, and is currently on antihypertensive drug therapy containing Enalapril, Hydrochlorothiazide, and Nifedipine. After few days of treatment, she came to the clinic with complaints of persistent cough and facial swelling. The blood pressure control was good and other findings were unremarkable.

What is the most appropriate drug therapy modification for this patient?

- (A) Replace Enalapril with Losartan
- (B) Replace Nifedipine with Amlodipine
- (C) Decrease the doses of all the medications
- (D) Replace Hydrochlorothiazide with Spironolactone

Answer: C

23. A 64-year-old man came with a poorly controlled blood pressure after two months of treatment with Hydrochlorothiazide and Losartan. The physician decided to add a third drug that works principally by reducing central sympathetic outflow.

What is the most likely medication prescribed by the physician?

- (A) Methyldopa
- (B) Captopril
- (C) Enalapril
- (D) Atenolol

Answer: A

24. A 62-year-old male patient with hypertension and chronic kidney disease was prescribed with hydrochlorothiazide 50 mg P.O. daily, Amlodipine 10 mg P.O. daily and Enalapril 10mg P.O. BID for one month. His physical examination shows blood pressure of 130/85mmHg and has grade two pitting edema in his extremities despite the current treatment.

What is the next most appropriate step in the management of this patient?

- (A) Increasing the dose of Enalapril
- (B) Replacing Amlodipine by Nifedipine
- (C) Increasing the dose of Hydrochlorothiazide
- (D) Replacing Hydrochlorothiazide by Furosemide

Answer: D

25. A 52-year-old female patient with diabetes, hypertension and chronic kidney disease came to a clinic for her regular follow up. Her home medications are Glibenclamide 5mg BID and Atenolol 25 mg P.O. daily. The attending physician approached a pharmacist to recommend another drug which will help for the reduction CKD progression.

What is the most appropriate drug to be added on her current treatment regimen?

- (A) Verapamil

26. A 62-year-old male patient with hypertension and chronic kidney disease was prescribed with hydrochlorothiazide 50 mg P.O. daily, Amlodipine 10 mg P.O. daily and Enalapril 10mg P.O. BID for one month. His physical examination shows blood pressure of 130/85mmHg and has grade two pitting edema in his extremities despite the current treatment.

What is the next most appropriate step in the management of this patient?

- (A) Increasing the dose of Enalapril
- (B) Replacing Amlodipine by Nifedipine

- (B) Hydrochlorothiazide
- (C) Hydralazine
- (D) Enalapril

Answer: D

A female patient with diabetes...
 26. A 55-year-old man with a history of hypertension for 10 years and intermittent seasonal allergic rhinitis since childhood came with complaints of persistent nasal itching, sneezing, clear rhinorrhea, and stuffiness with added ocular itching. His hypertension has been well controlled with Hydrochlorothiazide 25 mg every morning and Amlodipine 10 mg every day. He denies any other medical problems and his blood pressure is 120/80 mm Hg. He has no history of adverse drug reactions or drug allergies. He does not smoke and drink alcohol. He is a public bus driver.

What is the most likely recommendation to successfully control his seasonal symptoms?

- (A) Diphenhydramine
- (B) Clemastine
- (C) Cetirizine
- (D) Albuterol

Answer: C

27. A 25-year-old man with asthma was brought to a hospital with shortness of breath, chest tightness, wheezing and coughing at night. The physician ordered an investigation and the results showed a moderate persistent asthma.

Which is the most likely laboratory result consistent with the patient's condition?

- (A) FEV₁ is 30% of predicted
- (B) FEV₁ is 50% of predicted
- (C) FEV₁ is 75% of predicted
- (D) FEV₁ is 85% of predicted

Answer: C

28. A 36-year-old woman was presented to an OPD with complaints of heat intolerance and...
 Upon physical examination, the physician noticed a...
 On laboratory investigation, the thyroid hormones...
 and the diagnosis of Hyperthyroidism was...

Which technique that determined the physical findings?

- (A) Thyroid palpation
- (B) Thyroid Percussion
- (C) Thyroid Inspection
- (D) Thyroid Auscultation

Answer: A

most likely laboratory result

- (A) FEV₁ is 30% of predicted
- (B) FEV₁ is 50% of predicted
- (C) FEV₁ is 75% of predicted
- (D) FEV₁ is 85% of predicted

29. A 47-year-old woman with type II diabetes, who has been taking daily dose of Glyburide, came with increased thirst and excessive urination since one week after being put on some antihypertensive medication. Her plasma glucose was well controlled with the therapy until she had taken the antihypertensive medication. Her fasting blood glucose level currently is 150 mg/dL (normal range: 75-100 mg/dL).

What is the most likely antihypertensive drug she has been taking?

- (A) Captopril
- (B) Nifedipine
- (C) Methyldopa
- (D) Hydrochlorothiazide

Answer: D

30. A 42-year-old man with a chief complaint of severe lower abdominal pain after a falling down accident is admitted to the emergency department. He is a known asthma patient for the past 10 years that is controlled with β_2 -agonists and inhaled Beclometasone. He underwent an emergency laparotomy and a perforated appendix and associated peritonitis was revealed. Post-operatively, he became hypotensive and oliguric which does not respond well to IV dobutamine and despite extending the spectrum of his antibiotics. He remains hypotensive and becomes hypoglycemic. His blood chemistry shows Na^{++} 125 mEq/L, K^{+} 3.0 mg/dL. The physician diagnosed him as adrenal insufficiency with the following TSH and cortisol results.

What is the most appropriate treatment for the patient within 24 hours?

- (A) IV Hydrocortisone with 0.9% NaCl
- (B) IV Hydrocortisone with 0.9% NaCl
- (C) IV Hydrocortisone with 0.9% NaCl
- (D) IV Hydrocortisone with antibiotics

Answer: A

31. A 28-year-old woman with complaints of heat intolerance, weight loss, palpitations, and anxiety. A diffuse non-tender goiter is found and the thyroid function tests show low TSH and high T4 levels. She is diagnosed with

Graves' disease. A man with a chief complaint of severe lower abdominal pain after a falling down accident is admitted to the emergency department.

What is the most appropriate treatment for the patient within 24 hours?

- (A) Thyroidectomy
- (B) Propranolol
- (C) Propranolol
- (D) Propranolol

Answer: C

32. A 40-year-old male traffic controller at an Airport, comes to a hospital with complaints of indigestion and bloating of two-week duration. He is accompanied by indigestion and bloating of two-week duration. He is on a daily drug therapy of a proton pump inhibitor, usually between meals.

- (A) Hydrocortisone with 0.9% NaCl
- (B) Hydrocortisone with 0.9% NaCl
- (C) Hydrocortisone with 0.9% NaCl
- (D) Hydrocortisone with antibiotics

Answer: A

33. A 28-year-old pregnant woman comes with complaints of palpitations, oligomenorrhea, and anxiety. She is on a daily drug therapy of a proton pump inhibitor, usually between meals.

sometimes awakened him at night, but it has increased in frequency during the past week. Initially, the pain was temporarily reduced by food or antacids. Last week, he tried an OTC famotidine that "lasted longer" but did not provide adequate symptom relief. He used to smoke one pack of cigarettes daily for the past 20 years, has an occasional glass of red wine, and usually drinks 4 to 6 cups of caffeinated coffee throughout the day.

What is the most likely missed patient history in this case?

- (A) History of present illness
- (B) Past medical history
- (C) Medication history
- (D) Social history

Answer: B

33. A 65-year-old male patient has started treatment with parenteral iron therapy because he cannot tolerate an oral therapy for iron deficiency anemia. After three days, the patient developed anaphylaxis reaction.

What is the most likely responsible drug for occurrence of this reaction?

- (A) Iron dextran
- (B) Sodium ferrous gluconate
- (C) Iron sucrose
- (D) Ferrumoxytol

Answer: A

34. A 30-year-old man, who was on morphine for his moderate back pain, experienced severe diarrhea with abdominal pain, bleeding, fever, sepsis, electrolyte disturbances, and dehydration. The pain was episodic at first, exacerbated by meals, and somewhat relieved by defecation. He reports loss of appetite and a significant weight loss. Colonoscopy evaluations and pertinent laboratory investigations reveal a toxic megacolon secondary to *Clostridium difficile* infection. Finally, IV fluids, Metronidazole and Vancomycin were prescribed.

What is the LEAST likely important drug therapy in this patient?

- (A) Intravenous fluids
- (B) Metronidazole
- (C) Vancomycin
- (D) Morphine

Answer: D

35. A 40-year-old man, who was on morphine for his moderate back pain, experienced severe diarrhea with abdominal pain, bleeding, fever, sepsis, electrolyte disturbances, and dehydration. The pain was episodic at first, exacerbated by meals, and somewhat relieved by defecation. He reports loss of appetite and a significant weight loss. Colonoscopy evaluations and pertinent laboratory investigations reveal a toxic megacolon secondary to *Clostridium difficile* infection. Finally, IV fluids, Metronidazole and Vancomycin were prescribed.

What is the most likely responsible drug for occurrence of this reaction?

- (A) Intravenous fluids
- (B) Metronidazole
- (C) Vancomycin
- (D) Morphine

Answer: D

36. A 40-year-old man, who was on morphine for his moderate back pain, experienced severe diarrhea with abdominal pain, bleeding, fever, sepsis, electrolyte disturbances, and dehydration. The pain was episodic at first, exacerbated by meals, and somewhat relieved by defecation. He reports loss of appetite and a significant weight loss. Colonoscopy evaluations and pertinent laboratory investigations reveal a toxic megacolon secondary to *Clostridium difficile* infection. Finally, IV fluids, Metronidazole and Vancomycin were prescribed.

What is the best treatment option for this patient?

- (A) Eating small meals
- (B) Lansoprazole
- (C) Metronidazole
- (D) Vancomycin

Answer: C

37. A 40-year-old man, who was on morphine for his moderate back pain, experienced severe diarrhea with abdominal pain, bleeding, fever, sepsis, electrolyte disturbances, and dehydration. The pain was episodic at first, exacerbated by meals, and somewhat relieved by defecation. He reports loss of appetite and a significant weight loss. Colonoscopy evaluations and pertinent laboratory investigations reveal a toxic megacolon secondary to *Clostridium difficile* infection. Finally, IV fluids, Metronidazole and Vancomycin were prescribed.

What is the LEAST likely important drug therapy in this patient?

- (A) Intravenous fluids
- (B) Metronidazole
- (C) Vancomycin
- (D) Morphine

Answer: D

- (C) Anti-reflux surgery
- (D) Endoscopic therapies

Answer: B

36. A 34-year-old man came to a hospital with complaints of bleeding from mouth, headache, confusion, and ~~numbness~~ ~~in the limbs~~. He sent a laboratory test to confirm suspected Immune thrombocytopenic purpura (ITP). Some regular ~~laboratory~~ ~~test~~ ~~results~~ ~~are~~ ~~shown~~ ~~below~~.

What laboratory test ~~is~~ ~~most~~ ~~likely~~ ~~to~~ ~~be~~ ~~abnormal~~ ~~in~~ ~~this~~ ~~patient~~ ~~presentation~~?

- (A) Normal Prothrombin
- (B) Decreased reticulocyte count
- (C) Increased serum lactate dehydrogenase
- (D) Decreased lactate dehydrogenase

Answer: A

37. A 46-year-old hemophiliac patient is presented to the emergency department with complaints of joint pain, swelling, erythema, and mouth bleeding. After the patient is managed, the laboratory test revealed ~~normal~~ ~~results~~ ~~are~~ ~~shown~~ ~~below~~.

What could be the ~~most~~ ~~likely~~ ~~cause~~ ~~for~~ ~~this~~ ~~side~~ ~~effect~~?

- (A) Desmopressin
- (B) Tranexamic acid
- (C) Aminocaproic acid
- (D) Factor VIII replacement

Answer: A

38. A 50-year-old woman with the diagnosis of hemophilia and taken clotting factor replacement (ITP).

What is the ~~most~~ ~~likely~~ ~~cause~~ ~~for~~ ~~this~~ ~~patient~~?

- (A) Musculoskeletal trauma
- (B) Renal function test
- (C) Cardiac biomarkers
- (D) Thyroid function test

Answer: A

39. A 50-year-old hemophiliac patient is ~~presented~~ ~~with~~ ~~complaints~~ ~~of~~ ~~pain~~ ~~and~~ ~~stiffness~~ ~~in~~ ~~her~~ ~~right~~ ~~limb~~. The pain is ~~swelling~~ ~~erythema~~ ~~and~~ ~~lasts~~ ~~for~~ ~~about~~ ~~15~~ ~~to~~ ~~20~~ ~~minutes~~ ~~and~~ ~~then~~ ~~subsides~~ ~~she~~ ~~reports~~ ~~that~~ ~~she~~ ~~is~~ ~~having~~ ~~a~~ ~~difficulty~~ ~~of~~ ~~taking~~ ~~care~~ ~~of~~ ~~her~~ ~~self~~. She has ~~history~~ ~~for~~ ~~hypertension~~ ~~dyslipidemia~~ ~~and~~ ~~peptic~~ ~~ulcer~~. She could be the most likely cause for this ~~side~~ ~~effect~~. She was taking Hydroxychloroquine, Atorvastatin, and Omeprazole with good adherence.

What is the ~~most~~ ~~likely~~ ~~cause~~ ~~for~~ ~~this~~ ~~patient~~?

Answer: A

A 54-year-old woman has been presented with ~~complaints~~ ~~of~~ ~~pain~~ ~~and~~ ~~stiffness~~ ~~in~~ ~~her~~ ~~right~~ ~~limb~~.

The pain is ~~swelling~~ ~~erythema~~ ~~and~~ ~~lasts~~ ~~for~~ ~~about~~ ~~15~~ ~~to~~ ~~20~~ ~~minutes~~ ~~and~~ ~~then~~ ~~subsides~~ ~~she~~ ~~reports~~ ~~that~~ ~~she~~ ~~is~~ ~~having~~ ~~a~~ ~~difficulty~~ ~~of~~ ~~taking~~ ~~care~~ ~~of~~ ~~her~~ ~~self~~.

What is the appropriate management parameter?

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What is the appropriate management parameter?

- (A) Needs additional drug therapy
- (B) Unnecessary drug therapy
- (C) Ineffective drug therapy
- (D) Adverse drug reaction

Answer: A

40. A 37-year-old male patient with HIV/AIDS and on HAART was admitted to a hospital with generalized painful bullous lesion with accompanying high fever, headache, and respiratory symptoms that deteriorated rapidly. Lesions show rapid confluence and spread resulting in extensive epidermal detachment and sloughing. The clinical team diagnosed as Stevens-Johnson's syndrome (SJS) secondary to Cotrimoxazole as he was on it over the past 8 days. Therefore, they agreed to discontinue Cotrimoxazole immediately.

What is the next most appropriate drug therapy for this patient?

- (A) Aspirin
- (B) Ibuprofen
- (C) Diclofenac
- (D) Paracetamol

Answer: D

41. A 58-year-old man is diagnosed with colorectal cancer and came to a clinic complaining numbness and tingling in his right hand after eight cycle of his treatment.

What treatment is the most likely cause for this side effect?

- (A) Carboplatin
- (B) Leucovorin
- (C) Oxaliplatin
- (D) Capecitabine

Answer: C

42. About seven days after the first course of cytotoxic chemotherapy, a 49-year-old male patient experienced high grade fever. He has a profound neutropenia which is anticipated to extend 7 days. A history of an immediate-type hypersensitivity reaction from penicillin is reported. Despite an optimal dose of IV Ciprofloxacin and Vancomycin for his neutropenic fever management was instituted, the fever persisted. The physician requested the pharmacist to go through this case for a recommendation.

What should be done to achieve desired outcome for this patient?

- (A) Ciprofloxacin should be substituted by Gentamycin or Clindamycin
- (B) Fluconazole should be added to Ciprofloxacin and Vancomycin
- (C) Ciprofloxacin should be substituted by Ceftazidime or Cefepime
- (D) Acyclovir should be added to Ciprofloxacin and Vancomycin

Answer: B

43. A 50 kilogram weighing woman is being treated for her ovarian cancer by doxorubicin 45 mg/m² on day one, and Paclitaxel 200 mg/m² on day one. Currently the patient developed anemia and her hemoglobin increased by 0.5-1 g/dL after Epoetinalfa of 7500 units three times weekly is given for one month.

What is the most appropriate next step in the management of this patient?

- (A) Discontinue Epoetinalfa
- (B) Taper the dose of Epoetinalfa
- (C) Change Epoetinalfa to Darbepoetinalfa
- (D) Increase the dose of Epoetinalfa to 15000 units

Answer: D

44. A 41-year-old man with medical history of hypertension, type II diabetes, and cirrhosis comes to OPD with a complaint of erectile dysfunction. His current medication includes Enalapril, Metformin, Propranolol and Aspirin.

What drug is the most likely reason for this patient condition?

- (A) Aspirin
- (B) Enalapril
- (C) Metformin
- (D) Propranolol

Answer: D

45. A 20-year-old woman comes with a diagnosis of preterm labor. The gynecologist decided to postpone delivery long enough to allow for the maximum effect of antenatal steroid administration.

What is the most preferable drug for this patient?

- (A) Nifedipine
- (B) Terbutaline
- (C) Indometacin
- (D) Hydralazine

Answer: A

46. A 47-year-old man diagnosed with benign prostatic hyperplasia was presented with voiding symptoms and lower flow rate.

What is the most effective treatment option for this condition?

- (A) Surgery
- (B) Terazosin
- (C) Finasteride
- (D) Diphenhydramine

Answer: B

47. A 27-year-old male epileptic patient came to a pharmacy with an ordinary prescription of Phenobarbitone 100mg PO once daily for two months signed by a physician. The pharmacist on duty, after checking the date on the prescription was correct, provided the requested drugs with appropriate counseling for the patient and finally recorded the prescription.

What is the most likely error committed by the pharmacist?

- (A) Legality
- (B) Legibility
- (C) Completeness
- (D) Documentation

Answer: A

48. A mother of a five-year-old child, who complains of fever and poor appetite, went to a private pharmacy and made verbal request for Amoxicillin suspension. The pharmacist gave her the drug with an advice to shake it well and administer one teaspoon TID for ten days.

What was the most likely error of the dispenser in this case?

- (A) Shaking the bottle was not needed
- (B) He should have dispensed Augmentin instead
- (C) He should have asked her a prescription paper
- (D) Three times per day dose is not recommended for children

Answer: C

49. A 23-year-old man has started taking anti-depressant drugs prescribed by his physician. He has been taking the prescribed drugs as ordered. However, he came back to the hospital complaining constipation, dry mouth and difficulty in urination.

What drug is most likely responsible for the complaint of the patient?

- (A) Imipramine
- (B) Fluoxetine
- (C) Sertraline
- (D) Selegiline

Answer: A

50. A 34-year-old man was admitted to a hospital with a *Pseudomonas* urinary tract infection that is resistant to all antibiotics except for aminoglycosides.

What is the most appropriate strategy of using these drugs for this patient?

- (A) Initiating high dose of less frequently
- (B) Initiating high dose of more frequently
- (C) Initiating low dose of more frequently
- (D) Initiating low dose of less frequently

Answer: A

51. A 4-year-old child was brought to a hospital with a complaint of cough and fever. He was diagnosed with upper respiratory tract infection. The pediatrician prescribed two bottle of Azithromycin 200mg/5ml powder for suspension.

Which information should be given to the child's family?

- (A) Use suspension until it is finished
- (B) Use freshly boiled & cooled water(FBC)
- (C) Add water(FBC) to each bottle each time before using the drug
- (D) Add water (FBC) to the two bottles at the same time and keep safely

Answer: B

52. A prescription, containing Furosemide 20 mg PO BID and Spironolactone 100mg PO daily for one month, is presented to a hospital pharmacy. The available stock at hand for Furosemide is in 10mg strength and that of Spironolactone is 25 mg.

What is the total quantity of tablets to be dispensed for Furosemide and Spironolactone, respectively?

- (A) 240 tablets for both drugs
- (B) 120 tablets for both drugs
- (C) 120 and 160 tablets
- (D) 60 and 120 tablets

Answer: B

53. A 45-year-old man was diagnosed with type I diabetes mellitus. His fasting and post-meal blood glucose was 135mg/dl and 200 mg/dl respectively, HbA1C was 7.8% and BP was 130/82 mmHG. He was asymptomatic and investigations revealed no end organ damage. He was prescribed a treatment of: Lente Insulin 30IU a.m /20IU p.m for 120 days. (Hint: U-100 insulin contains 100 units/mL and 1vial=10ml).

How many vials should be dispensed for this patient?

- (A) 4
- (B) 6
- (C) 12
- (D) 3

Answer: B

54. A 60-year-old man was presented for the third time gouty attack in the past one year. He feels pain on his left knee and the third and fourth proximal interphalangeal joints of his left hand. He has a history of peptic ulcer. His pain started about 10 hours ago and rates his pain as 6/10.

Which OTC is most appropriate for treatment of this acute gouty attack?

- (A) Indomethacin

- (B) Diclofenac
- (C) Piroxicam
- (D) Celecoxib

Answer: D

55. An extemporaneous prescription order calls to prepare a mix of 2 % (w/v) and give 1tsp QID PO ac. A pharmacy professional prepared it by taking 50ml of a 4%w/v stock solution and 50 ml of the appropriate diluents. While dispensing the pharmacist labeled as to take 5ml every 6 hours by mouth before a meal.

What is the most likely error committed by the pharmacist in this case?

- (A) Volume of stock solution used
- (B) Volume of diluents used
- (C) Amount used during preparation
- (D) Labeling during dispensing

Answer: D

56. A Pharmacist received a prescription to prepare Bismuth Subgallate Suppositories BP. The pharmacy professional prepared the medication with the right SOP procedure and labeled it accordingly.

What precaution is required during labeling this preparation?

- (A) For external use only
- (B) Warm before use
- (C) For rectal use only
- (D) Store in a cool place

Answer: C

57. A 40-year-old male pharmacist is interested to ascertain the quality of pharmaceutical products in store pharmacy by simple physical inspection.

What is the most appropriate indicator for good quality pharmaceuticals in this inspection?

- (A) Ripped packaging
- (B) Non-crumpled pills
- (C) Unreadable labels
- (D) Moisture inside injectable package

Answer: B

58. A 33-year-old pregnant woman visited a pharmacy with a prescription paper. She gave the prescription paper to a male pharmacist who wore a clean and white gown. He attached his identification tag on the gown in such a way that it was visible to the client. He dispensed the prescription correctly while he was talking his girlfriend via telephone.

What is the error of the above dispensing practice?

- (A) Patients lack reception
 - (B) Prescription was illegal
 - (C) Legibility of the prescription
 - (D) Dispensary room was not well organized
- Answer: A

59. A 31-year-old dispensary pharmacist has planned to arrange all the pharmaceutical products on a newly made shelf for ease of dispensing.

What method of drug arrangement should be avoided in this scenario?

- (A) Dosage forms
 - (B) Pharmacotherapeutic category
 - (C) Alphabetical order by brand name
 - (D) Alphabetical order by generic name
- Answer: C

60. A 30-year-old pregnant woman was taking drugs for motion sickness. According to animal study, the drug she was taking causes some abnormality on fetus and the mother. But no human study has been done yet on the drug.

What is the appropriate FDA pregnancy category of the above drug?

- (A) A
 - (B) B
 - (C) C
 - (D) D
- Answer: B

61. A 50-year-old obese woman, who was diagnosed with Type-II Diabetes mellitus 15 years back and on regular NPH insulin, came to an outpatient pharmacy for an advice regarding the side effects of her anti-diabetic drugs. She complains blurring of vision, weakness, headache, anxiety, confusion, increased appetite and diaphoresis immediately after taking the drug.

What could be the most appropriate advice for the patient in this case?

- (A) Taking only NPH Insulin
 - (B) Taking only regular Insulin
 - (C) Switching back to oral agents
 - (D) Taking sugar containing meal
- Answer: D

62. A 56-year-old male asymptomatic patient came to OPD and vital sign showed a BP of 180/110 mmHg. He was not previously treated for hypertension and the physician put him on Nifedipine slow release 30 mg PO daily.

What is the main reason for selecting slow release preparation for this patient?

- (A) To increase efficacy
- (B) To increase adherence
- (C) To decrease side effect
- (D) To decrease total daily dose

Answer: C

63. A 66-year-old alcohol addicted male patient faced a repeated severe headache for the last two weeks. A neighbor tells him that he had similar problem previously and got a relief after taking frequent Paracetamol tablet. The neighbor shares the anti-pain for the patient and after three days, the patient is admitted to a hospital and diagnosed with paracetamol-induced hepatitis.

What factor is mainly responsible for the hospitalization of this patient?

- (A) Because the patient was sharing a medicine
- (B) Because the patient was using ineffective drug
- (C) Because the patient was using inappropriate medicine
- (D) Because the patient is old, he is susceptible to hepatitis

Answer: A

64. A mother comes to a pharmacy to buy an OTC medication to treat dry cough of her 6 month infant. The pharmacist dispensed dextromethorphan hydrobromide for her infant. The mother came back to the pharmacy the next day and complained of worsened condition of her baby.

What is the main reason for the client's complaint?

- (A) The infant must be put on antibiotics and dextromethorphan
- (B) The infant must be on optimal dose of chlorpheniramine syrup
- (C) The infant must be referred to a hospital from the beginning
- (D) The infant must be put on codeine phosphate rather than dextromethorphan

Answer: C

65. In Good Dispensing Practice, following a counseling session, transactions related to the prescriptions must be recorded on the prescription registration format to monitor performances. In addition, each prescription should be signed by an authorized dispenser and filled prescription should be kept as a receipt.

What activity is proper with regard to documentation and prescription filling?

- (A) Special prescriptions should be filed for five years
- (B) Prescriptions should be disposed carefully by the pharmacy staff
- (C) Prescriptions records should be accessible to authorized prescribers
- (D) Prescriptions should be collected monthly and filled chronologically

Answer: A

66. A junior pharmacist at a district hospital was asked by a nurse for advice on the type of medical supplies which can be sterilized and reused.

Which type of medical supplies can most likely be sterilized for reuse?

- (A) Disposable syringe
- (B) Degradable sutures
- (C) Glass syringes
- (D) Expired syringes

Answer: C

67. A medical director of a certain hospital requested twenty blood pressure measurement devices from a pharmacy store. The pharmacist prepared all the blood pressure measurement devices and packed.

What is the most appropriate next step in dispensing these devices?

- (A) Supply the devices
- (B) Evaluate the request
- (C) Request to bring a bill
- (D) Check the repackaging

Answer: A

68. A pharmacist working in a store is requested for nasogastric tube (NG) via Report and Requisition Form (RRF).

What should the store pharmacist consider before issuing the requested item?

- (A) Size
- (B) Length
- (C) Cost
- (D) Thickness

Answer: A

69. A 48-year-old female patient tells a drug information service pharmacist that she saw a story on the internet that state as patients should stop taking their warfarin and switch to another agent. She tells the pharmacist that she is not planning to take it anymore and wants the physician to find her an option. She requested the pharmacist to recommend her any other alternatives.

What is the most likely approach that the pharmacist will follow to help this client?

- (A) He should provide appropriate responses irrespective of the reliability of internet
- (B) He should advise patient to refrain from seeing internet as an extension of care
- (C) He should review and evaluate the information before recommendation
- (D) He should refer the patient to her physician for decision

Answer: C

70. A clinical pharmacist working at an emergency department observed a significant number of children with diabetic ketoacidosis (DKA). On the pharmaceutical assessment report, the pharmacist realized that most patients were injecting their insulin on the same site as it is less painful. He wants to prepare drug information material for these patients.

What would be the prioritized topic of interest for those patients on insulin use?

- (A) Dosing and preparations
- (B) Administration technique
- (C) Related adverse events
- (D) Stability at room temperature

Answer: B

71. Drug information query dealing about the dose of Vancomycin for the treatment of a patient admitted with bacterial meningitis is submitted to a drug information center. A pharmacist assigned to the center wants to obtain the relevant information to formulate effective response.

What is the most relevant background information required to solve the above problem?

- (A) Sex of the patient
- (B) Creatinine clearance
- (C) Route of administration
- (D) Address of the patient

Answer: B

72. A randomized clinical trial was conducted to assess the effect of aspirin in stroke patients for about two years. The study showed a significant mortality difference at six months (5% in aspirin versus 10% in placebo). At two years, risk of fatal bleeding due to aspirin was 12% compared to 0.3% in placebo. The mortality rate at two years was 15% in aspirin and 15% in placebo.

What would be the appropriately synthesized recommendation from this study?

- (A) Long term use of aspirin in stroke patient is very safe
- (B) Long term use of aspirin in stroke patient increases mortality
- (C) Long term use of aspirin in stroke patient reduces mortality
- (D) Long term use of aspirin in stroke patient is not beneficial

Answer: D

73. A pharmacist planned to conduct searching of literatures regarding the effectiveness of coffee in prevention of liver cancer on secondary databases.

What would be the appropriate search strategy to obtain relevant literatures on the topic?

- (A) Coffee AND effectiveness

- (B) Coffee OR effectiveness
- (C) Coffee OR liver cancer
- (D) Coffee AND liver cancer

Answer: D

74. A pharmacist planned to evaluate randomized clinical trial published on the efficacy of Warfarin and Dabigatran among individuals who have undergone a knee surgery. The pharmacist is interested to identify information on the comparative finding of this research with other similar studies.

Which article section will give such kinds of information?

- (A) Abstract
- (B) Conclusion
- (C) Discussion
- (D) Results

Answer: C

75. A group of scientists working on drug discovery and development from plant source performed size reduction of the plant extract before extraction. The next step in the process is to isolate the active constituents from the dried and powdered plant. They require obtaining a more concentrated extract by soaking the extract and flushing with a fresh solvent.

What is the best extraction procedure they should employ for this purpose?

- (A) Distillation
- (B) Maceration
- (C) Percolation
- (D) Counter-current extraction

Answer: C

76. A Pharmaceutical technologist who is working in a research and development unit is planning to prepare a dosage form, with a fast onset of action, for emergency treatment.

Which tablet formulation is more suitable for this situation?

- (A) Chewable
- (B) Enteric coated
- (C) Effervescent
- (D) Super disintegrating

Answer: C